

Project Report On
“REAL-TIME HELMET DETECTION SYSTEM USING
COMPUTER VISION AND ESP32”

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CERTIFICATE

This is to certify that,

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have satisfactorily completed the Project entitled, **“REAL-TIME HELMET DETECTION SYSTEM USING COMPUTER VISION AND ESP32”**. This work is being submitted in partial fulfillment for the Second Year in Artificial Intelligence and Data Science of the Shivaji University, Kolhapur, Maharashtra, INDIA for the academic year 2025-2026.

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Abstract

The Helmet Detection System is a real-time application developed using Python, computer vision, and ESP32 to enhance road safety by automatically detecting whether a person is wearing a helmet. The system captures live video through a webcam and uses a YOLO-based object detection model to identify persons. It then applies image processing techniques such as colour detection and blur analysis to determine the presence of a helmet.

Based on the detection result, signals are sent to the ESP32 microcontroller, which controls LEDs to indicate the status: green for helmet detected, red for no helmet, and yellow for no person detected. The system works in real time, reduces manual monitoring, and improves rule enforcement.

Overall, this project provides a low-cost and efficient solution for traffic monitoring and serves as a foundation for future smart traffic management systems.

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1.INTRODUCTION

1.1 Project Idea:

The project titled *Real-Time Helmet Detection System Using Computer Vision and ESP32* is designed to automatically detect whether a two-wheeler rider is wearing a helmet or not. The system captures live video using a webcam and processes each frame using computer vision techniques. The main goal is to create an automated solution that can help in monitoring and enforcing traffic safety rules without the need for continuous human supervision.

The system uses the OpenCV library to perform image processing and object detection tasks. It detects the rider's face using the Haar Cascade Algorithm and then analyses the region above the head to identify the presence of a helmet. Once detection is completed, the result is sent to the ESP32, which controls LED indicators to display whether a helmet is detected or not.

1.2 Need of the Project:

Road accidents involving two-wheeler riders are increasing rapidly, and one of the major reasons for severe injuries is the absence of helmets. Despite strict traffic rules, many riders neglect wearing helmets due to carelessness or lack of enforcement. Manual monitoring by traffic police is not always effective, especially in crowded areas and during peak hours.

This project addresses the need for an automated and reliable system that can continuously monitor helmet usage. By using computer vision, the system reduces human effort and ensures real-time detection. It can be deployed at traffic signals, highways, and other important locations to improve road safety and minimize accidents.

1.3 Literature survey:

- In recent years, various researchers have worked on helmet detection systems using computer vision and artificial intelligence techniques. These studies focus on improving road safety by automating the detection of helmet usage.
- (P. Viola & M. Jones, 2001) This study introduced the Haar Cascade algorithm for rapid object detection. It is widely used for real-time face detection due to its speed and efficiency. In helmet detection systems, it helps in identifying the face region, which is further used to detect the presence of a helmet. However, its accuracy decreases in complex environments and varying lighting conditions.
- (Joseph Redmon et al., 2016) This research presented the YOLO (You Only Look Once) object detection model, which provides high accuracy and fast detection performance. YOLO is widely used in helmet detection systems for identifying riders and helmets in real time. However, it requires high computational power and GPU support, making it less suitable for low-cost systems.
- (Szeliski, 2010) This study explains image processing techniques including color detection using the HSV (Hue, Saturation, Value) model. HSV-based methods are commonly used in helmet detection to identify helmet colors. These methods are simple and fast but are sensitive to lighting conditions and color variations.
- (Espressif Systems, 2020) This research focuses on the ESP32 microcontroller used in IoT applications. It enables real-time communication between software and hardware components. In helmet detection systems, ESP32 is used to control LEDs or signals based on detection results. It is cost-effective and suitable for embedded systems.

2. PROBLEM STATEMENT & SCOPE

2.1 Problem Statement:

Many riders do not wear helmets, and manual checking of helmet usage is not efficient or practical at all locations. Therefore, there is a need to develop an automated system that can detect whether a rider is wearing a helmet or not. The system should be capable of working in real-time, should be low-cost, and must be able to control hardware components such as traffic signals based on the detection result.

2.2 Scope:

- Implementation at traffic signals and toll plazas.
- Use in college campus entry monitoring.
- Real-time webcam-based detection.
- Focus on low-cost hardware.
- Accuracy depends on lighting and helmet color.
- Future upgrade using YOLO/CNN.

2.3 Area of Project:

This project belongs to multiple technological domains, including computer vision, artificial intelligence, embedded systems, and IoT applications. It integrates software and hardware components to create a smart and automated detection system.

2.4 Goals & Objectives:

The main goal of this project is to develop a system that can detect a rider's face using the Haar Cascade algorithm and identify the presence of a helmet using HSV colour detection techniques. The system is designed to send the detection result to the ESP32 microcontroller, which then controls LEDs based on whether a helmet is detected or not. Another important objective is to develop a cost-effective and efficient system that can be easily implemented in real-world scenarios.

3. SOFTWARE REQUIREMEN SPECIFICATION

3.1 Software Requirements:

Software Requirement	Specification
Python	Used for system programming
OpenCV	Used for image processing.
NumPy	Used for numerical operations.
Arduino IDE	Used to program ESP32.

Table 3.1 Software Requirements

3.2 Hardware Requirements:

Hardware Requirement	Specification
ESP32	Controls LEDs.
Webcam	Captures video.
LEDs	Show output indication
Laptop	Runs the system.

Table3.2 Hardware Requirements

4.PROJECT PLAN

4.1 Project schedule

Month	Week 1	Week 2	Week 3	Week 4	Week 5
January	Introduction to mini project	Select project topic	Study literature survey	Define problem and scope	Prepare synopsis
February	Start project setup	Design of project	Implement face detection	Implement helmet detection	-
March	Improvement in Code and features	Final testing	Complete project report	Complete documentation	Finalize and submit project

5.SOFTWARE DESIGN

5.1 Flow chart:

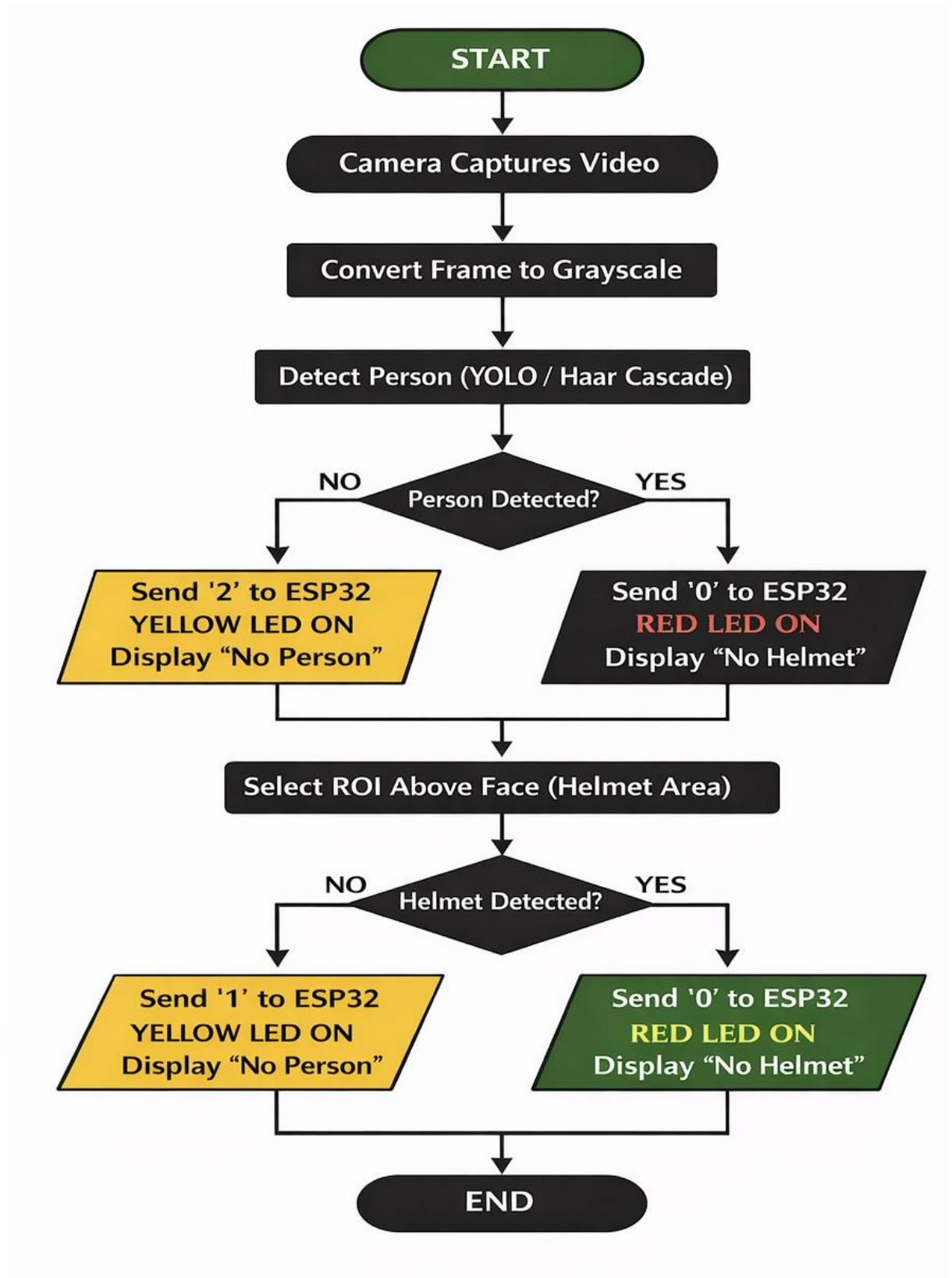


Fig. 5.1 Flowchart of Real-Time Helmet Detection System with ESP32

6. IMPLEMENTATION DETAILS

6.1 Modules and Their Functionalities:

- **Video Capture Module:**

The video capture module is responsible for capturing real-time video using a webcam. It continuously collects image frames from the camera, which are then forwarded to the processing modules for further analysis.

- **Face Detection Module:**

The face detection module uses the Haar Cascade algorithm to identify human faces in the captured video frames. It detects the position of the rider's face, which helps in locating the region where the helmet is expected.

- **Helmet Detection Module:**

The helmet detection module uses HSV (Hue, Saturation, Value) colour detection to identify the presence of a helmet. It analyses the region above the detected face and checks for helmet-like colour patterns to determine whether a helmet is present.

- **Decision Module:**

The decision module processes the output from the helmet detection module and determines whether the rider is wearing a helmet or not. Based on this decision, it generates the appropriate result for further action.

- **Communication Module:**

The communication module is responsible for sending the detection result from the computer to the ESP32 microcontroller using serial communication. It transmits signals such as '1' for helmet detected and '0' for no helmet.

- **Hardware Module:**

The hardware module includes the ESP32 and LEDs, which act as output devices. Based on the received signal, the ESP32 controls the LEDs, where the green LED indicates helmet detection and the red LED indicates no helmet.

7. Testing

7.1 INTRODUCTION:

The helmet detection system is tested to ensure accurate detection of helmets using real-time video processing and communication with ESP32. The testing focuses on verifying detection logic, system performance, and hardware response under different conditions such as lighting, motion, and object variation.

1. Functional Testing:

Functional testing is performed to verify that each part of the system works correctly. The webcam is tested for proper video capture, the YOLO model is tested for detecting persons, and the helmet detection logic (blur and colour ratio) is checked. The system correctly identifies helmet and no-helmet conditions and displays appropriate messages.

2. Detection Accuracy Testing:

The system is tested with different scenarios such as riders wearing helmets, not wearing helmets, and partially visible helmets. The blur value and colour ratio conditions are verified to ensure accurate classification. The system performs well when helmet colour is dark and clearly visible.

3. Real-Time Performance Testing:

The system is tested in real-time using continuous video input. It ensures that frames are processed without delay and detection results are displayed smoothly. The YOLO model processes frame efficiently, providing near real-time performance.

4. Noise Handling Testing:

The helmet counter and NOISE_LIMIT are tested to reduce false detection. The system avoids sudden changes in output by stabilizing results over multiple frames. This improves reliability and prevents flickering between helmet and no-helmet states.

5. Communication Testing:

Serial communication between Python and ESP32 is tested. The system sends '1' for helmet, 0' for no helmet, and '2' for no person detected. It ensures that data is transmitted correctly and only when the state changes.

6. Hardware Testing:

The ESP32 and LEDs are tested to verify correct output. The green LED turns ON when a helmet is detected, the red LED turns ON when no helmet is

detected, and the yellow LED indicates processing. The hardware responds accurately to the received signals. testing is conducted under real-time conditions.

7.2 TYPES OF TESTING:

1. Unit Testing:

Each module such as video capture, YOLO detection, blur calculation, and colour detection is tested individually to ensure correct functionality.

2. Integration Testing:

All modules are combined and tested together to verify proper interaction between video processing, detection logic, and ESP32 communication.

3. Performance Testing:

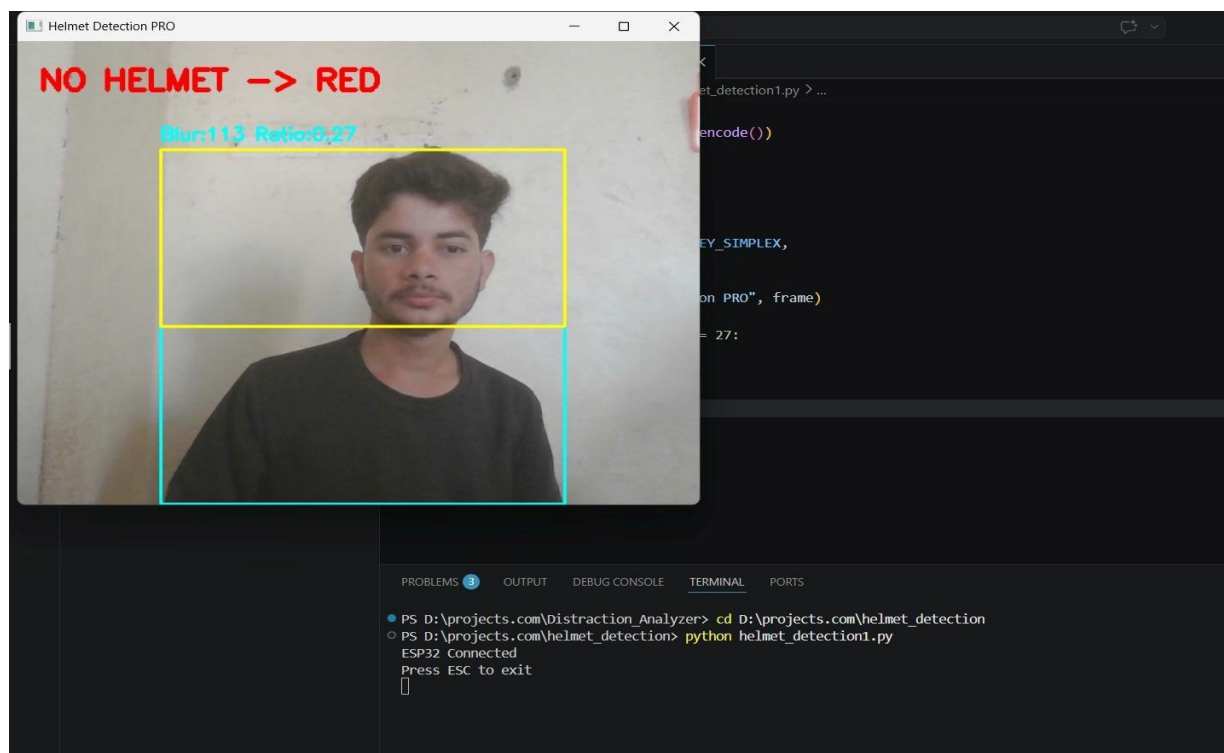
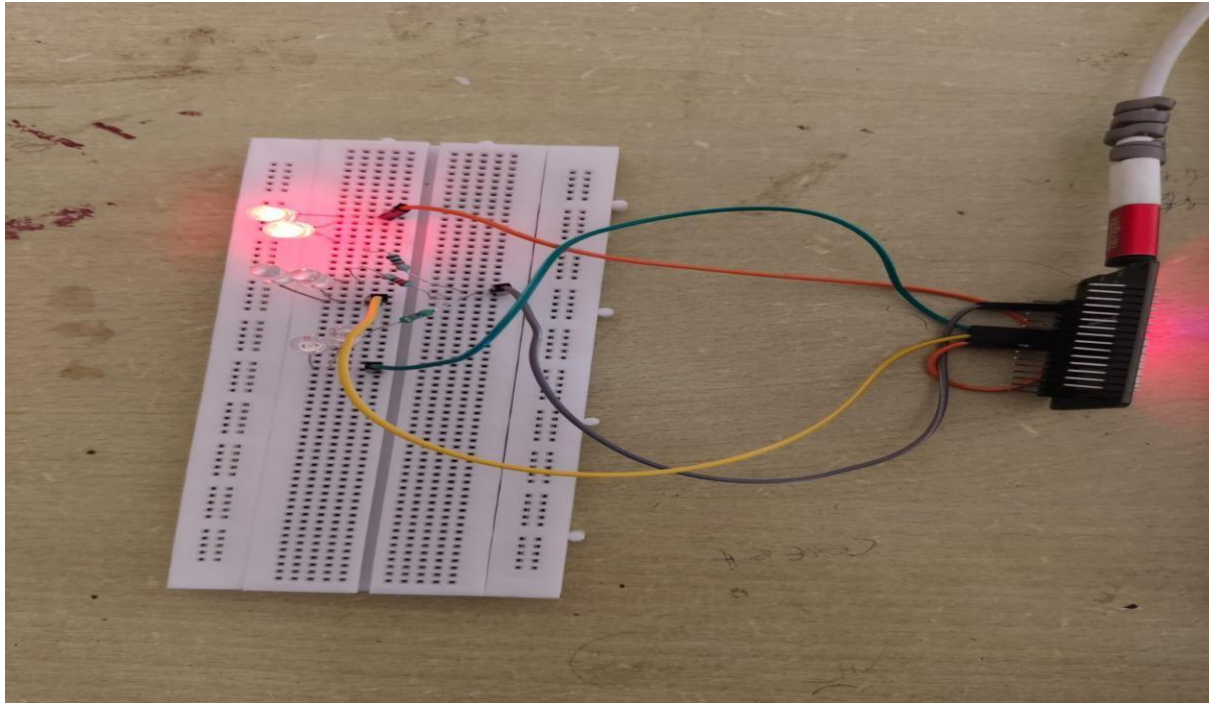
The system is evaluated for speed and efficiency in real-time conditions to ensure smooth operation without lag.

4. Hardware Testing:

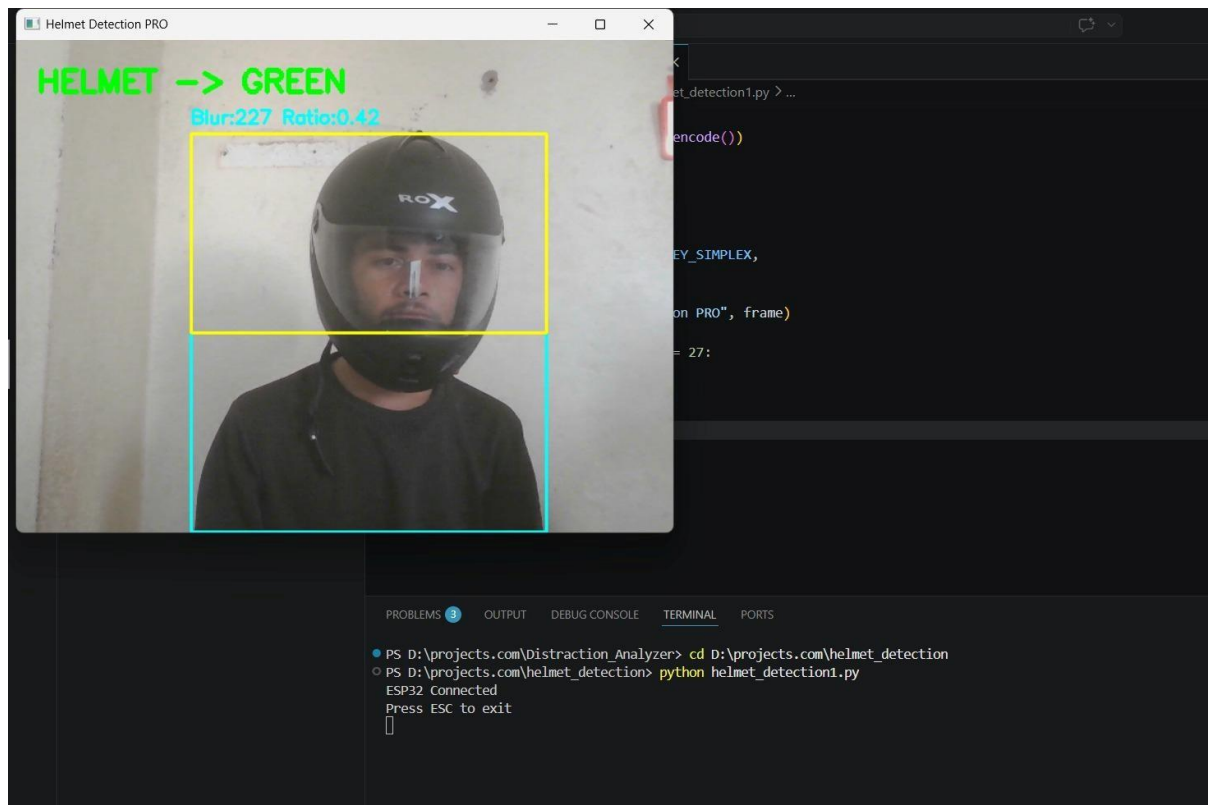
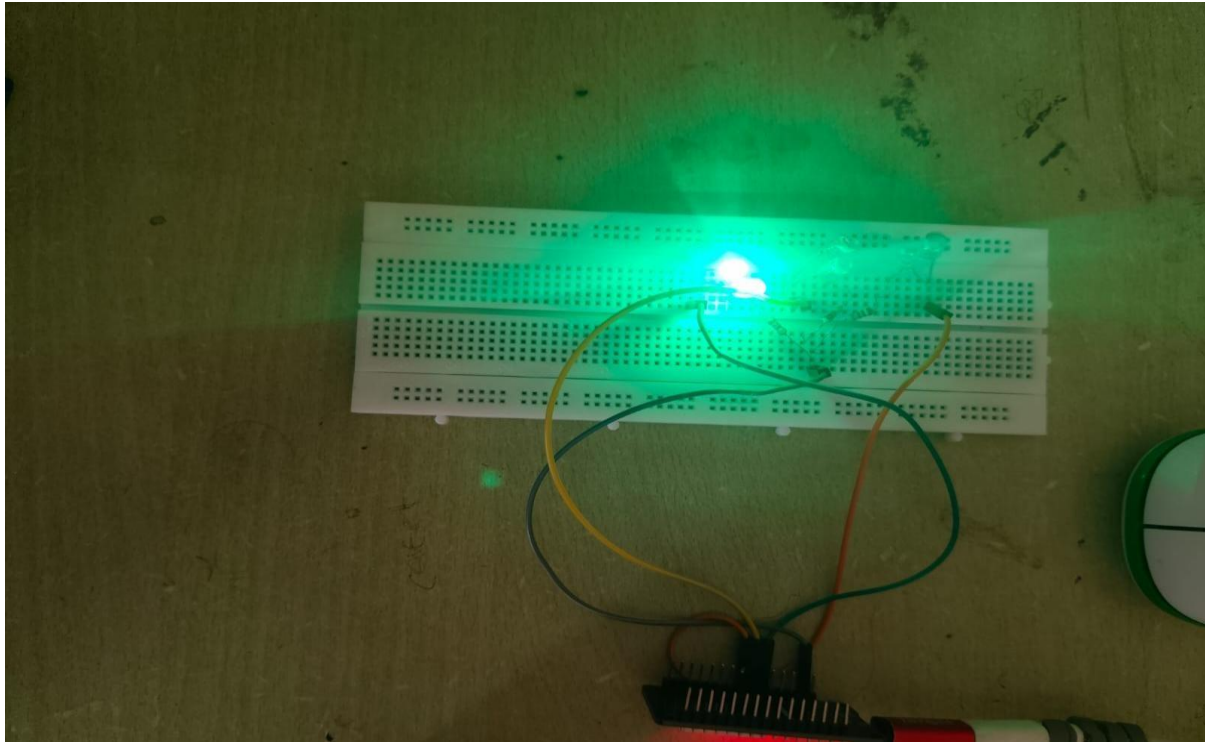
The ESP32 and LEDs are tested to confirm proper response to signals sent from the Python program.

8. SNAPSHOTS/ GUI

- **No helmet detected**



- Helmet detected



9. CONCLUSION

The Real-Time Helmet Detection System was successfully developed using computer vision techniques and hardware integration. The system effectively detects whether a rider is wearing a helmet by analyzing live video input using the YOLO model and image processing methods. Based on the detection result, it communicates with the ESP32 microcontroller to control LEDs, providing a clear indication of helmet usage.

The project demonstrates a low-cost, efficient, and real-time solution for improving road safety and reducing manual effort in traffic monitoring. The implementation of noise handling and decision logic ensures stable and accurate results under different conditions. The system also includes a no person detection state, indicated by a yellow LED, which improves accuracy and avoids unnecessary processing.

Overall, the system achieves its objective of automatic helmet detection and can be practically used in traffic signals, college entrances, and surveillance systems. It also provides a strong foundation for future improvements using advanced artificial intelligence and automation technologies.

10. FUTURE SCOPE

The Real-Time Helmet Detection System can be further improved and enhanced using advanced technologies and additional features. In the future, deep learning models such as YOLO (You Only Look Once) and Convolutional Neural Networks (CNN) can be used to increase detection accuracy and handle complex real-world conditions more effectively.

The system can be integrated with CCTV cameras for large-scale traffic monitoring across cities. It can also be connected to cloud platforms to store data, generate reports, and enable remote monitoring. Automatic challan (fine) generation systems can be added to enforce traffic rules without human intervention.

Further improvements can include license plate recognition to identify violators, mobile application integration for real-time alerts, and IoT-based smart traffic systems. The system can also be enhanced to work effectively in low-light conditions and detect helmets of different color and designs.

Overall, the project has strong potential to evolve into a smart traffic management system that improves road safety, reduces accidents, and supports automated law enforcement.

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